

A multi-graph learning framework to fuse heterogeneous market information for stock forecasting

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ABSTRACT

Traditional financial studies focused on the fundamental indicators data and financial news to track stock movements. Recent studies have revealed that firm relationships, measuring spillovers from related firms, are also important for stock forecasting. However, previous studies via machine learning mostly relied on incomplete market information and failed to capture the interactions and heterogeneity within multimodal data. In this study, a multi-graph neural network framework (MGNN) is proposed to fuse heterogeneous market information with different data structures, i.e., numerical, textual and graphical representations. With such a design, the unique and heterogeneous nature of multimodal data is well addressed. Specially, a fine-grained event extraction methodology is employed to identify valuable events within a substantial corpus of financial news, thereby ensuring the reliability and quality of the extracted events. Then, an explicit and an implicit graph are constructed to integrate firm relationships with fundamental and news information, respectively. Finally, the distinct spillover effects of fundamental and news information are further integrated using a multi-scale cross-attention dual-branch transformer to capture the joint impact on stock movements. Experiments performed on real market data demonstrate the effectiveness of the proposed approach over state-of-the-art algorithms.

1. Introduction

Tracking stock movements is key for investors making decisions and seeking to maximize profits in the stock market. It is also an important work part of financial institutions and is often involved in the economic decision-making process. However, the stock market can be highly volatile, non-stationary and noisy, which makes predicting stock prices a great challenge. In fact, stock market is a complex dynamic system in which various influencing factors intertwine and interact with each other to guide its movement. Previous studies have roughly attributed stock fluctuations to three types of market information: fundamentals (e.g., turnover, opening prices, trading volumes), financial news and company relation (Huang et al., 2022b; Tan et al., 2022). Therefore, the problem of modeling stock movements is essentially a multimodal learning problem.

To capture their joint impact, a number of machine-learning based approaches are proposed to incorporate various types of information from multimodal big data, rather than being limited to a single type of information in traditional econometric models, such as auto regression integrated moving average (ARIMA) (Ariyo et al., 2014) and generalized autoregressive conditional heteroskedasticity (GARCH) (Herwartz, 2017). For example, Ding et al., extracted semantic event information from news articles with a deep learning framework and fed them, along with historical stock prices, into a tensor neural network to forecast stock trends (Ding et al., 2016). Feng et al., adopted graph learning approach to incorporate company relationships and consider the diffusion of stock historical information in the company graph for stock prediction (Feng et al., 2019). Unfortunately, these studies that used such a fusion strategy have incorporated only some types of market information, which ignored the function of certain modal data and to some ex-

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tent failed to mitigate the stochastic nature of stock price fluctuations (Wang et al., 2021).

In addition, multimodal data exhibit distinct data structures and encompasses a multitude of data formats (Li et al., 2021a). Most of the existing studies have not focused on this heterogeneity of multimodal data, simplifying data processing during the fusion stage. Specifically, fundamental information typically consists of numerical indicators, while financial news is expressed in textual form and company relationships are often represented in graphical data formats. For textual news, unlike numerical indicators for fundamental information, this information needs to be quantized into scalars or vectors, which are then fused with fundamental indicators through a concatenation operation for stock forecasting. A common strategy is to obtain scalar indicators by counting the number of news or the number of sentiment words, or to use word embedding methods to obtain representation vectors for the text (Li et al., 2016; Tetlock et al., 2008). This counting method inevitably results in an inherent loss of substantial contextual information within the content. While word embedding techniques offer a potential remedy to mitigate this loss, they are characterized by a degree of generality that may encompass extraneous and noisy information, which is not beneficial for models to learn complex decisions.

For the graphical relation, it is used to connect firm nodes to simulate the information diffusion in the stock market, which enables the convergence and aggregation of spillover effects. Previous studies mainly relied on the explicit relations, such as the industry category or shareholders (Chen et al., 2018; Feng et al., 2019; Xu et al., 2021b), to construct a firm graph and fused fundamentals and financial news as node representations to capture the influence of related firms. In fact, the fundamentals of a firm are affected by its relevant firms known as momentum spillovers, while the impact of financial news on a firm is strengthened or weakened by news impact of its related firms known as media spillovers (Tan et al., 2022). Utilizing such a fusion schema that initially integrates features at the firm node level and subsequently performs relational fusion presents challenges in effectively capturing the spillovers inherent in the different information modalities. Furthermore, these linked relations should vary across information in different modalities. Apparently, fusing fundamental and news information at the node level in the first stage lacks the relational diversity and versatility needed to distinguish between different spillovers.

To address the above challenges, we propose a novel multi-graph neural network framework (MGNN) to fuse multimodal data for tracking stock movements. Essentially, the proposed fusion framework shed lights on addressing multimodal information in the stock market, that is a typical challenge in fusing diverse information to alleviate volatility stochasticity. Although previous studies have demonstrated the importance of three types of market information in stock volatility, it still lacks an efficient way to adequately integrate them into stock forecasting. To the best of our knowledge, this is a pilot study to apply a multi-graph learning framework to fuse multimodal market factors represented in different data structures (numerical, textual and graphical) and to explore their joint effects and spillovers on stock fluctuations. The study makes the following three unique contributions.

- To preserve high-quality information in financial news, a fine-grained event extraction approach is adopted to filter out distracting information and locate valuable events related to specific companies;
- To distinguish spillovers from data of different modalities, we constructed multiple graphs in which an explicit relation is used for momentum spillovers in fundamentals and an implicit relation based on the extracted event triples is employed for media spillovers in financial news;
- To adequately handle multimodal data, we propose a cross-attention dual-branch transformer that further fuses fundamental and news information, which have been fused with relational data, at different time scales to capture their joint impact on stock movements.

In this study, all modules are seamlessly integrated into the proposed fusion framework in an end-to-end manner. This mechanism allows for simultaneous integration at both the firm level and the inter-firm level. Experiments performed on real stock markets demonstrate the superiority of the proposed approach over state-of-the-art algorithms, including LSTM, GCN, RGCN, MFNN and MAC. In particular, the proposed framework achieved statistical enhancements of at least 0.39% and 1.40% in terms of DA and AUC, respectively.

2. Related Work

This study is closely related to two major directions of research: media-aware stock movements, and deep learning for multimodal data fusion. In this section, we focus on reviewing the relevant literature from these two perspectives.

2.1. Media-aware Stock Movements

In behavioral finance studies, stock movements are attributed to cognitive and affective biases of investors (Mitchell & Mulherin, 1994), that is, the release, dissemination, and absorption of news information can lead to fluctuations in the stock market. The pilot study on media-aware stock movements can be traced back to work on the effects of financial report on stocks (De Bondt & Thaler, 1985). Essentially, this media effect behind news reports is regarded as a leading indicator of media information (Jiao et al., 2020), and its effect can be continuously enhanced by repetition and discussion on other social media. Later, some scholars observed the impact of emotions extracted from financial news (Schumaker & Chen, 2008; Wang et al., 2012). In particular, investors' decisions can be influenced by the opinions of others as expressed via media news, which may result in herd behaviors in stock market. For example, Si et al. (2014) calculated the public sentiment of a stock based on tweets about the stock itself and related stocks, and found that such sentiment indicator could predict stock volatility. Considering the limited information contained in a single numerical sentiment indicator, some studies took a step further by employing embedding vectors to represent textual news. For instance, Ma et al. (2023) used a pre-trained sentiment classifier to generate market-driven sentiment embeddings from news headlines, and demonstrated the predictive effectiveness of such embedding representations for stocks. He et al. (2022) used the Word2Vec and the dictionary method to measure and aggregate textual tone in the media news as the media-based investor sentiment index. Huang et al. (2022a) applied the embedding method ELMo to represent news titles, rather than the entire news content, to eliminate the effect of noisy text on stock predictions.

However, these counting and embedding methods locate too condensed or too generalized information, respectively, making it difficult to focus on valuable information for tracking stock movements. In essence, financial news is usually lengthy, with only a few specific elements attracting the attention of investors and causing stock movements (Cheng et al., 2020; Wu & Wang, 2021). Cheng et al. (2020) found that extracting structured event triples from textual news articles can better capture stock movement for making financial investments. Further, some studies have demonstrated that stock prediction tasks can benefit from extracting event SPO triples (subject, predicate, object) that generally contain sufficient and comprehensive information to summarize the important content of financial news. In these studies, some simple heuristic rules like SPO have been used to establish connections between words, event triples, and sentences. However, the textual information of financial news presents a complex syntax in reality, and extracting value events through certain simple rules inevitably results in the loss of valid information. In this study, we propose a fine-grained event extraction method with several useful triple rules that can be designed to filter out distracting information and identify meaningful events related to a specific company, especially some financial textual elements. With

this design, it allows for the conversion of textual data into a machine-friendly form and explores its predictive power in media-aware stock prediction.

2.2. Deep learning for multimodal data fusion

Another related research topic is the study on the deep learning for multimodal data fusion. As aforementioned, the stock market is a complex dynamic system in which various market factors mingle together to cause stock fluctuations. The multimodal information can be broadly categorized into three types, including numerical fundamentals, textual financial news, and graphical company relationships (Huang et al., 2022b). Inspired by the research in computer science (Wu & Han, 2018; Xu et al., 2019; Zadeh et al., 2017), some studies focused on fusing multimodal data through deep learning to explore the joint role of various market information in tracking stock movements. For example, Li et al. (2016) adopted a tensor-based schema to fuse economic indicators and news sentiment and demonstrated its predictive effects. Zhang et al. (2018) fused fundamental data and financial news articles via a coupled matrix and tensor factorization framework and found that capturing their joint effects leads to a better sense of stock movements. Chen et al. (2022) proposed a self-adaptive gate layer to perform stock prediction by fusing numerical market information and social text information. However, these studies ignore graphical company relationships and failed to adequately capture spillover effects in stock market, such as momentum and media spillovers.

To further fuse company relationships, some scholars have utilized graph neural networks, leveraging their abilities of aggregating and transmitting information in the constructed graphs (Kipf & Welling, 2016). Specifically, pioneering studies define the entire market as a graph where each node represents an individual firm, and edges between nodes are determined by company relationships, such as industry (Feng et al., 2019), shareholders (Chen et al., 2018), or price movement (Li et al., 2021b). For example, Chen et al. (2018) built a graph for listed companies based on financial investment facts and then explored stock movements via GCN. Lin et al. (2018) proposed a graph convolutional feature based convolutional neural network (GC-CNN) to forecast stock trend. Later, Xu et al. (2021b) proposed a novel relational event-driven forecasting framework (REST) to address the propagation of event spillovers from related stocks. Cheng et al. (2022) proposed a multi-modality graph neural network framework (MTGNN) to incorporate complex spillovers in the market for forecasting.

However, these graph-based studies fuse relationships with fundamental or news information, which ignores certain types of information. In addition, even considering all three types of information, they fuse fundamentals and news at the firm node level, followed by relational fusion, which fails to distinguish between the spillovers inherent in different information modes (Xu et al., 2021a). For example, Zhang et al. (2023) combined transaction data and textual data into a node vector and applied the attributes-driven graph attention network for stock predictions. As stated in the work of (Tan et al., 2022), such information messaging forces all spillover effects to be subservient to the same relationships and inevitably fails to distinguish the spillover paths of media spillover and momentum spillover. Furthermore, some studies have found that the handling of different temporal scales of modalities is also important in fusion schemes due to the heterogeneity of different modes of multimodal information. For example, Liu et al. (2020) proposed a multi-scale two-way deep neural network (MTDNN), and found that learning multi-scale patterns from two types of market information can better track stock volatility.

To address the above issues, in this study, we adopt a multi-graph learning framework to fuse multimodal market factors represented in different data structures (numerical, textual and graphical), and attempt to simultaneously integrate multimodal information at both the firm level and the inter-firm level to fully consider their heterogeneity.

3. Preliminaries

Before providing details of our proposed model, we briefly introduce some formal definitions in this study.

1) *Problem Definition*: The aim of this paper is to learn a fusion model M . This model learns information representations of stock s_i on the trading day t through multimodal data, including the historical event information e_i^t , the fundamental information of historical stock data f_i^t and the relation information r_i between stocks in the firm graph G , to predict the stock price trend of day $t + 1$. Specifically,

$$trend_i^{t+1} = M(e_i^t, f_i^t, r_i | \theta) \quad (1)$$

where θ represents a set of parameters in the proposed fusion framework.

2) *Features and Labels*: Here, the historical event information e_i^t is generated by extracting event triplets from firm-specific financial news during the past 10 trading days. Then the event triplets are processed and represented as event information vectors that can be further used as inputs in the learning process. The fundamental information f_i^t consists of historical stock trading data in the last 10 trading days, including the opening price, closing price, highest price, lowest price and volume. The relation information can be seen as an undirected graph $G = (V, E)$, where V denote a set of stocks in the market and E is the set of relationships between two stocks. A represents the adjacent matrix of stock graph G . $A_{i,j} = 1$ means that there is an existing relation between stock s_i and stock s_j while $A_{i,j} = 0$ indicates that stock s_i and stock s_j are not related. Stock price trend forecasting is essentially a binary problem, i.e., predicting the stock price trend for each stock on the next trading day $t + 1$. Similar to many previous studies, we define change rate of the closing price $Price_{close}$ as the stock price trends (labels). For stock s_i , the label on i^{th} day can be calculated as,

$$y_i^t = \begin{cases} \text{upward} & r_i^t \geq 0 \\ \text{downward} & r_i^t < 0 \end{cases} \quad (2)$$

$$r_i^t = \frac{price_{close_i}^{t+1} - price_{close_i}^t}{price_{close_i}^t} \quad (3)$$

4. Methodology

In this study, we propose a multi-graph fusion framework to simultaneously address multiple types (numerical, textual, and graphical) of market information to capture stock movements, each of which has been proven to be effective in shaping stock trends in financial research. Fig. 1 is an overview of the proposed framework.

4.1. News event extraction

Financial news articles contain rich disclosures and their release, dissemination and absorption can lead to stock movements. Previous studies have mainly focused on sentiment words, or embedding the entire news content to represent textual news for tracking stock movements. These methods may identify information that is somewhat limited or contains redundancies, and thus might not fully reflect or capture the inherent influence of news articles (Ding et al., 2015; Duan et al., 2018). Therefore, this study proposes a fine-grained event extraction approach to locate firm-specific events in the financial news.

1) *Event Triple Extraction*: Considering the complex syntax in financial news, this study attempts to represent news using event triples. An event triple can be expressed as $e = (s, p, o)$ where p is the action or predictive, s is the actor or subject and o is the object on which the action is performed. Such SPO triples can easily coalesce information throughout a sentence and are often seen as ideal representations of information in news articles. Therefore, to extract event triples, we first adopt a Chinese dependency parsing tool, DDParse (Zhang et al., 2020), to perform dependency analysis on financial news to obtain a syntactic dependency

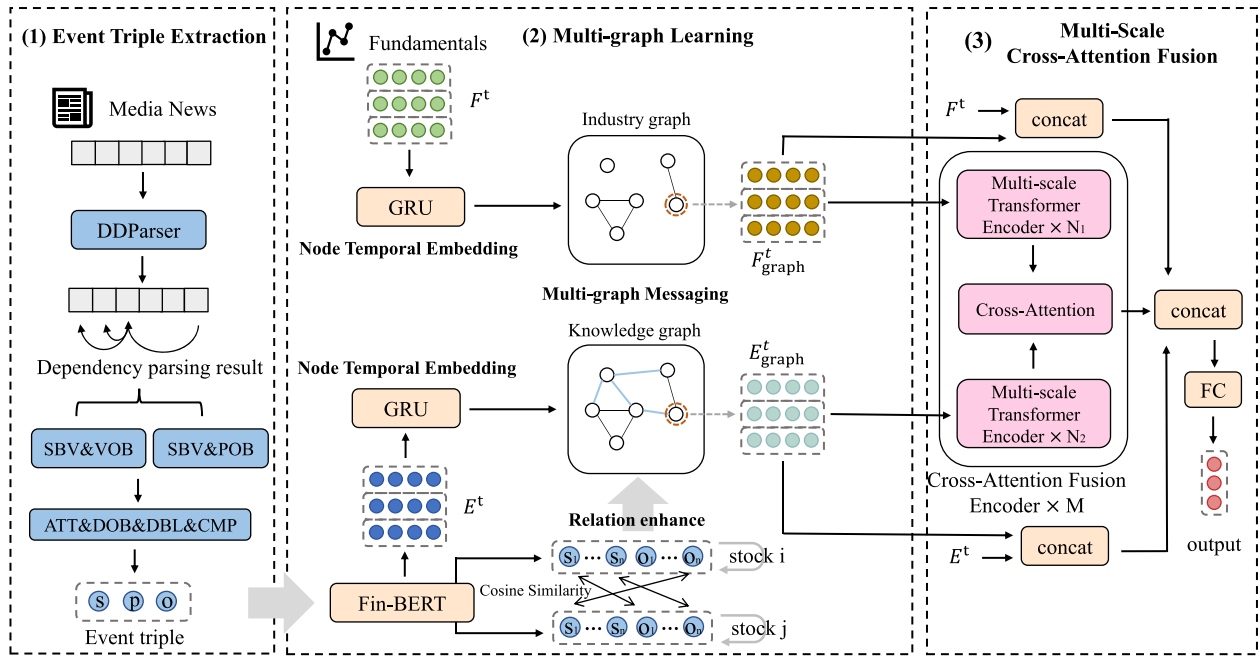


Fig. 1. The overview of the proposed MGNN framework.

tree of sentences in the news texts. After obtaining the syntactic dependency tree, we create a dictionary to store the dependency parsing relation rules of every word in each sentence. Then, we use two classic rules in terms of the action or predictive (p) to extract event triples. In particular, the first rule is a combination of the subject-verb (SBV) and verb-object (VOB). Based on this rule, the most common sentence patterns can be extracted to form the event SPO triples. The second rule is the combination of the subject-verb (SBV) and preposition-object (POB). The SBV relationship first finds subjects and predicates. The POB relationship is further used to recognize prepositions and objects and assigns prepositions to predicates. After applying these two rules, the event SPO triples can be obtained in terms of the constructed syntactic dependency tree.

However, in the real market, financial news articles present a complex syntax, and extracting event triples through relatively simple syntactic rules may lead to the loss of some valid information. For instance, a news related firm A reports that firm A sells the shares of firm B to firm C. The traditional method focuses on finding the direct subject, predict and object through SBV, VOB and POB rules, and can only extract the event triple ($A, sell, shares$), which not only omits entity firm B, but also ignores the relationship between firm A and firm C. It is obvious that relying on subject-verb-object and subject-verb-object rules alone is not sufficient to extract useful SPO even triples. Here, we provide in-depth insights into financial news contents and collect suggestions from some expert analysts to summarize several specific event extraction rules as supplements to the classical SPO extraction principles. Specifically,

- ATT: represents the relation between an attribute and the head noun it modifies. For example, “shares of firm B” includes ATT relation (*firm B, shares*).
- DOB: represents the relation between the predict and the direct and indirect objects. For example, “firm A sells the shares of firm B to firm C” includes DOB relation (*sell, shares of firm B*) and (*sell, firm C*).
- DBL: represents the relation between the object and the verb that follows. For example, “firm A forced its employees to redundant” includes DBL relation (*its employees, redundant*).
- CMP: represents the relation between the predict and its complementary components. For example, “firm A was established in 2018” includes CMP relation (*establish, 2018*).

By adding these syntactic rules, more high-quality firm-specific events can be extracted. In addition, it offers a higher probability of discovering some hidden relationships between listed companies. The extracted event triples enable financial news to represent firms and model their relationships by capturing specific economic events that correspond directly to core phenomena in finance. This structured format defines the nature of inter-firm links, offering a more precise channel for analyzing spillover effects compared to simple co-occurrence or sentiment methods.

2) *Event Embedding*: To represent financial news as firm node embeddings, the extracted event triples need to be further encoded as vectors. Considering that the event triples are extracted from financial news and contain many words related to the financial field, the event triples are vectorized using Fin-BERT (Araci, 2019). Specifically, the subject s , predicate p and object o of each event triple are encoded via Fin-BERT and output as a 768-dimensional vector, respectively. On each trading day, the firm-specific news articles related to a target firm may contain several event triples. Thus, the embeddings of s , p and o can be regarded as average vectors of these triples. Based on comparative experiments where averaging demonstrated superior and more robust performance than other aggregation methods, the three mean vectors ($\bar{s}, \bar{p}, \bar{o}$) are used to form the mean-event triple matrix H , which comprehensively represents the news text information regarding the target stock. The embedding of news event triples H can be indicated as,

$$H = (\bar{s}, \bar{p}, \bar{o}) = \frac{1}{n} \sum_{i=1}^n (s_i, p_i, o_i) \quad (4)$$

Here, the dimension size of each average event triple matrix H is 768×3 . Then, matrix H is flattened into a vector h of length 2304. The obtained vector h can be considered as the embedding representation of the event information for the target stock on a trading day. The high-dimensional flattened vector features exhibit sparsity primarily due to irregular occurrence of news events, which results in numerous zero values in the inputs. We further incorporate an autoencoder to reduce the dimensionality of the event embedding. The autoencoder employs a structure of two fully connected layers, that is, a linear layer and an activation layer, which can be defined as,

$$e_i^t = \text{LeakyReLU}(\text{Linear}(h)) \quad (5)$$

After projection through the autoencoder, the dimension of e_i^t is relatively small and suitable for further processing in subsequent graph learning. These parameters are estimated in the learning task simultaneously. Finally, the obtained vector e_i^t represents the news event embedding of stock i on day t .

4.2. Multi-graph learning

Typically, stock fluctuations are attributed to three categories of stock market information, including fundamentals, financial news and firm relationships. Recent successful stock prediction methods are very adept at capturing hidden patterns in complex multimodal data (Huang et al., 2022b; Tan et al., 2024). In this study, we propose a multi-graph learning mechanism to integrate multimodal market information and enable the relational diversity and versatility needed to distinguish between different spillovers.

1) *Multiple Graph Construction*: To distinguish different spillovers, we constructed multiple graphs in which the industry graph with explicit relation is used for momentum spillovers in fundamentals, and the knowledge graph with implicit relation based on the extracted event triples is employed for media spillovers in financial news.

We first construct a stock graph G_1 where nodes represent firms, and edges for momentum spillovers are determined by real-world industry relationships. The adjacency matrix $A_1 \in R^{N \times N}$ represents G_1 , with N as the firm count. Using event triples extracted from financial news, we adjust stock linkages to enhance media spillovers by identifying related firms through subject-object pairings from target stock news. Direct relationships are detected by searching entities in event triples that match manually collected firm names (official and alternative names of listed companies), updating A_1 to A_2 by incrementing $a_{i,j}$ for each connection.

Additionally, to capture indirect relationships via shared affiliates or similar event topics, we link firms based on event triple similarity. The intuition is that firms conveying similar information likely exhibit event correlations and spillover effects. With fine-grained event extraction, triples contain ample firm-related or event-related entities. As shown in Figure. 1, we list all the subjects and objects of the target stock (i.e., firm i), through which we calculate the cosine similarity with another potentially relevant stock (i.e., firm j). If it exceeds threshold α , a relationship is established, with weight increasing for more matching entities, denoted as,

$$W_{ij} = \sum_{m=1}^M \sum_{n=1}^N \mathbb{I}(S(x_m, y_n) > \alpha) \quad (6)$$

$$S(x_m, y_n) = \frac{x_m \cdot y_n}{\|x_m\| \|y_n\|} \quad (7)$$

Here, w_{ij} is the weight of the relationship between firm i and firm j . $M(N)$ denotes the number of entities in the subject and object list of target firm i (firm j). x_m and y_n represent the entity embedding of firm i and firm j , respectively. $S(\cdot)$ is the cosine function used to compute the cosine similarity between two entity embeddings. Details of the calculation of the cosine function are presented in Equation (7). Finally, we update the adjacent matrix A_2 by adding the weight w_{ij} to $a_{i,j}$, which combines both the direct and indirect linkages of the two firms. Similar to graph G_1 , the nodes in G_2 represent listed firms and the edge between the two nodes for media spillover effects are determined by the enhanced firm relationships. That is, stock graph G_2 can be considered as an enhanced graph with the knowledge of news events added to graph G_1 . With this design, multiple graphs can be constructed to distinguish different spillover effects, thus providing diverse information messaging for multimodal data.

2) *Node Temporal Embedding*: As mentioned above, stock graph G_1 is constructed for momentum spillovers, which allows for the fusion of fundamentals and firm relationships, while stock graph G_2 is designed for media spillover effects, which considers the fusion of news events and firm relationships. In this study, although both graphs G_1 and G_2 have firms as graph nodes, the embedding of the nodes are fundamental

information and news information, respectively. In particular, vector f_i^t denotes the fundamental information of firm i on day t , and its dimension represents the number of fundamental features including opening price, highest price, trading volume, turnover and so forth. Vector e_i^t represents the event embedding of firm i on day t , as described in Section 4.1.

However, the stock market is a dynamic system and market signals are time-series. This suggests that the current state of a firm is influenced by its previous state, i.e., historical state. To capture such time dependence, this study uses the sequential data during the past T days as embedding vectors for the firm nodes in graphs G_1 and G_2 . Specifically, a single-layer GRU is adopted to capture time-series characteristics, as it is efficient enough in practice, denoted as,

$$\tilde{e}_i^t = GRU(e_i^t, e_i^{t-1}, \dots, e_i^{t-T}) \quad (8)$$

$$\tilde{f}_i^t = GRU(f_i^t, f_i^{t-1}, \dots, f_i^{t-T}) \quad (9)$$

Here, the final sequential embeddings $\tilde{f}_i^t$ and $\tilde{e}_i^t$ are the last hidden state of GRU. These two vectors contain not only the current state information on day t (f_i^t, e_i^t), but also the historical information during the past days, i.e., $F_i^{[t-T:t]} = (f_i^{t-1}, \dots, f_i^{t-T})$, $E_i^{[t-T:t]} = (e_i^{t-1}, \dots, e_i^{t-T})$. More details of GRU can be found in the study of Cho et al. (2014).

3) *Multi-graph Messaging*: In the market, stock fluctuations of a focal firm are affected not only by its historical information but also by the impact of its related firms, which is known as spillovers in finance. Depending on the type of information diffusion, such spillover effects mainly involve momentum spillover and media spillover. We propose a multi-graph learning approach to consider different spillover mechanisms in multimodal data and capture the spillover effects based on the constructed graphs. For a target stock, we aggregate information on its related firms in parallel in the constructed graphs G_1 and G_2 to capture momentum spillovers and media spillovers, respectively. In graph G_1 , the aggregation process aims to merge fundamental information of the neighboring nodes of a target node, while in graph G_2 , it combines the news event information. To achieve this, we utilize a two-layer graph convolutional network (GCN) to aggregate node features in the stock graph and generate embeddings. The parameters of GCN are not shared between graphs G_1 and G_2 , so different spillover mechanisms can be implemented. Thus, for graph G_i , the nodes feature matrix of the l th layer in GCN can be denoted as,

$$H_i^{(l+1)} = \sigma(\tilde{D}_i^{-\frac{1}{2}} A_i \tilde{D}_i^{-\frac{1}{2}} H_i^{(l)} W^{(l)}) \quad (10)$$

where $H_i^{(l)} (i = 1, 2)$ represents the nodes feature matrix of the l th layer in GCN. D_i is a diagonal degree matrix which is used to normalize the adjacent matrix A_i . $W^{(l)}$ is the weight matrix to be estimated in the learning process, and $D_{ii} = \sum_j A_{ij}$. $\sigma(\cdot)$ is the activation function. ReLU activation function is adopted in this study. Specially, the initial node feature matrix can be denoted as $H_1^{(0)} = F^t$ and $H_2^{(0)} = E^t$. Considering the complexity of multimodal data, we further adopt a Dropout layer to reduce the overfitting phenomenon by eliminating some of the input features. The Dropout layer of $H_i^{(l+1)}$ can be denoted as,

$$H_i^{(l+1)} = Dropout(H_i^{(l+1)}) \quad (11)$$

After the two-layer GCN architecture, the outputs F_{graph}^t and E_{graph}^t can be obtained from graphs G_1 and G_2 , respectively. For a target stock i , its embedding in graph G_1 , is the i th raw data of the output F_{graph}^t , which contains the fundamental information spillover of its neighboring stocks at the industry level. Its embedding in graph G_2 , is the i th raw data of the output E_{graph}^t , which contains the news information spillover of its neighboring stocks at the event knowledge level. With this design, we can fuse fundamentals and financial news with firm relationships separately and obtain their respective spillover impacts in terms of momentum spillovers and media spillovers.

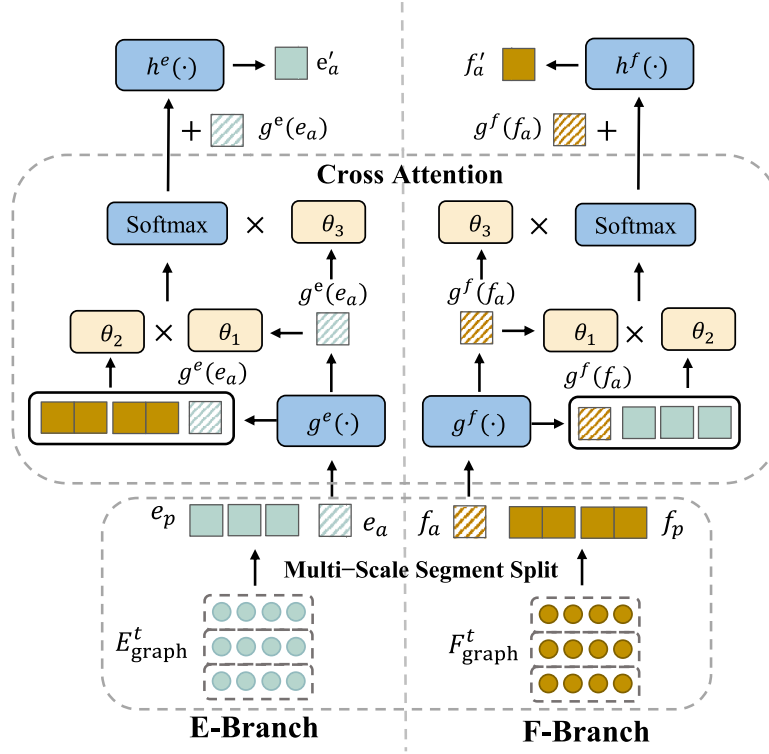


Fig. 2. Multi-scale cross-attention fusion for multimodal market information.

4.3. Multi-scale cross-attention fusion

As mentioned above, most of the deep learning-based approaches focus on only partial types of multimodal market information. Even those considering all types of modality commonly rely on simple feature concatenation. Thus, after obtaining F_{graph}^t and E_{graph}^t , an appropriate schema is needed to fuse the embedding of these two different modalities to capture their joint impact on stock movements. To get better representations of the fusion information, we propose a dual-branch network architecture to further integrate the fundamental and news information that have been already fused with relational information, respectively. Fig. 2 illustrates the multi-scale cross-attention fusion architecture of two-modal market information.

Considering that E_{graph}^t obtained from event extraction contains more semantic and boundary information, while F_{graph}^t contains more explicit information, we design two different branches (E-Branch and F-Branch) in the encoder for these two types of multimodal data. Specially, inspired by the study of CrossViT (Chen et al., 2021), we adopt different scales to split the information in the two branches for subsequent interactive fusion. For E-Branch, it has higher dimensional features, so it is split into smaller scale segments e_p to retain more embedding information with more transform encoders. For F-Branch, which is much lower dimensional and can be divided into larger scale f_p , it is possible to represent this modal information efficiently using only fewer encoders.

To fuse E_{graph}^t and F_{graph}^t more efficiently and effectively, we establish the information interaction between e_p and f_p . Fig. 2 illustrates the cross-attention fusion for E-Branch and F-Branch. E_{graph}^t and F_{graph}^t are first passed through two transformer encoders with different sizes N_1 and N_2 . Then, an additional token is added to the embedding E_{graph}^t and F_{graph}^t , which is similar to the function of classification token in BERT (Devlin et al., 2018) to gather all the fusion information together. e_p and f_p represent additional tokens of the input embedding of E-Branch and F-Branch, respectively. In training, the embedding of the additional token is initialized randomly. e_a and f_a is actually an agent of one branch with the function of exchanging information of f_p and e_p from the other

branch and then back to its own branch. Specially, the procedure of e_p exchanging information of f_p from F-branch is as follows:

$$e' = \text{concat}(g^e(e_a), f_p) \quad (12)$$

where $g^e(\cdot)$ is the projection function for dimension alignment. We then integrate the multimodal information using a cross-attention module between e_a and e' , where e_a only serves to fuse the information from e' into itself. Specifically, the cross-attention module can be expressed as

$$\text{Attention} = \text{softmax}((e_a \theta_1)(e' \theta_2)^T / \sqrt{d/h}) \quad (13)$$

$$\text{Cross-Attention} = \text{Attention} \cdot (e' \theta_3) \quad (14)$$

As did in self-Attention, we also use multiple heads in the Cross-Attention. Specially, d and h are the e_a 's dimension and number of heads, respectively. $\theta_i (i = \{1, 2, 3\})$ is the learnable parameters of each part. Like CrossViT, there is no Multi-Layer Perceptron (MLP) module after the Cross-Attention embedding fusion. We utilize another projection function $h^e(\cdot)$ and a shortcut block to avoid gradient vanishing. The output e'_a can be expressed as

$$e'_a = h^e(\text{Cross-Attention}(l(e')) + g^e(e_a)) \quad (15)$$

where $l(\cdot)$ is layer normalization which is aim to improve the training efficiency. Similarly, F-branch follows the same procedure but the object to exchange is the opposite. Both branches are fused together for the prediction task, and the last e_a and f_a are further concatenated and used as the fusion feature of E_{graph}^t and F_{graph}^t . Then we concatenate all the information as the matrix X^t :

$$X^t = \text{concat}(e'_a, f'_a, E_{graph}^t, F_{graph}^t, E^t, F^t) \quad (16)$$

After obtaining X^t , we feed X_i^t (the i th row of X^t) representing fused information of stock s_i , to a FC layer with the softmax function to generate the probability of future stock trends for stock s_i on day $t+1$, denoted as,

$$\hat{y}^t = \text{softmax}(W^T X^t + b) \quad (17)$$

where W is the weight matrix and b is the bias vector. By minimizing the binary cross-entropy loss between the predicted result $\hat{y}^t$ and ground truth y^t , the parameters of the proposed framework can be learned.

5. Experiments

To the best of our knowledge, this study is the first attempt to explore a multi-graph fusion framework to address multimodal market information for tracking stock movements. To gauge the effectiveness and robustness of the proposed framework, we conducted a series of experiments from the perspective of predictability.

5.1. Experiment settings

1) *Datasets*: Although stock transaction data is accessible, there are few public stock datasets containing multimodal market information, including numerical, textual and graphical data for stock price forecasting. In this study, a multimodal real-word dataset is collected to measure the efficiency and generalization of the proposed framework. In particular, this dataset contains the market transaction data of Chinese A-share market and B-share market from December 30, 2017 to January 1, 2019, and the financial news crawled from 12 mainstream financial news websites in China. The news text underwent standard preprocessing. This included removing HTML tags, and eliminating non-alphanumeric characters and extra whitespace. There are 1,295 stocks with 439,037 news articles during the period of 243 transaction days. Note that, this study focuses on multimodal market information, thus we further collect the industry firm relationship from the Chinese Stock Market & Accounting Research (CSMAR) database, covering the selected 1,295 stocks during the same trading period mentioned above. As for the implicit event relevant relationship, we also public it along with our constructed multimodal dataset. To test performance across market sizes, we built firm graphs at two scales: one with the full dataset (large-scale) and another with 300 random stocks (small-scale). In order to guarantee the robustness of the evaluation, the experiment was repeated 10 times for each random selection of small-scale data.

In addition, our dataset is divided sequentially into three periods. The first 80% of transaction days are used for training, the following 10% of days for validation, and the last 10% of the days for testing. The evaluation is conducted in a rolling-window fashion, as suggested by (Cheng & Li, 2021a). That is, the market signals during the past T transaction days were utilized to predict future stock trends. To ensure the robustness of the evaluation, for each comparison, we trained it 10 times with different initializations. As suggested by (Feng et al., 2019), the average performance of these runs in the testing period is reported to eliminate fluctuations caused by different initializations.

2) *Evaluation Metrics*: Most studies of tracking stock movements via machine learning treat the prediction task as a binary classification problem. Only a few researches regard it as a regression task (Li et al., 2017). To make the proposed framework comparable, this study also treated this as a binary classification problem. Specifically, if the closing price on the next day is higher than the closing price today, the sample is labeled with “upward” ($y_i = 1$), otherwise labeled with “downward” ($y_i = -1$). To gauge the performance of the proposed framework, three most popular metrics adopted in the previous studies (Li et al., 2021a; Xu et al., 2021a) are utilized as evaluation metrics in this study, namely, directional accuracy (DA), F1 and area under the curve (AUC). Specifically,

$$DA = \frac{TP + TN}{TP + TN + FN + FP}, \quad (18)$$

$$AUC = \int_0^1 \frac{TP}{TP + FN} d \frac{FP}{FP + TN}, \quad (19)$$

$$F1 = \frac{2TP}{2TP + FN + FP} \quad (20)$$

where $TP(TN)$ denotes the number of positive (negative) samples correctly classified as positive (negative). $FP(FN)$ is the number of negative (positive) samples falsely classified as positive (negative) samples. The DA and AUC are widely adopted as the evaluation metrics for stock classification tasks. DA measures the difference between predicted

Table 1

The parameter setting of our model.

Parameter	Explanation	Value
max_norm	max norm of the gradients	0.25
lr	learning rate	0.001
dropout	dropout rate	0.2
hid_gru	hidden size of GRU layer	64
out_channel	out channel of GCN layer	16
heads	heads in Cross-Attention module	3
ca_depth	depth of Cross-Attention module	1

trends and actual changes in stock prices, and AUC obtained from the receiver operating characteristic (ROC) curve depicts the area under the ROC curve. For both metrics, a larger value indicates better predictive performance of the model.

3) *Parameter Settings*: The basic framework of the proposed method is implemented in Pytorch. Grid search is applied to tune the hyper-parameters to achieve the optimal performance. In this study, window size T for capturing time-series characteristics (in Eq. 8 and Eq. 9) is searched within $\{5, 10, 15, \dots, 50\}$. Different scales (in Section 4.3) are searched in the range of $\{2, 4, 8, 16\}$. For the proposed neural network, we test different layers and dimensions within $\{8, 16, 32, 64, 128, 256\}$. Different cosine similarity thresholds α (in Section 4.2) are searched in the range of $\{0.2, 0.5, 0.7, 0.8, 0.9\}$. In our preliminary experiments, the optimal window size T is set to 10. The scales for E-Branch and F-Branch are set to 4 and 2, respectively. Fig. 3 present the prediction performance in terms of DA and AUC with different T , the event scale, and fundamental scale for better understanding. The hidden dimensions of GRU module and graph neural network are set to 16 and 32, respectively. All network parameters are initialized using Glorot (Glorot & Bengio, 2010) and trained using the Adam optimizer (Kingma & Ba, 2015) with an initial learning rate of 0.001. We train a maximum of 100 epochs. Each epoch cost about 1200 seconds. We conducted all the experiments on a Nvidia GeForce RTX 4090 GPU. Table 1 summarizes the basic parameters in our model.

5.2. Comparison

To evaluate the overall performance of the proposed framework, we compared it with several baselines, including five classic and four state-of-the-art models. For baselines, we use their default recommended parameters. In particular,

1) *Classic Classifiers*: Classical classifiers are capable of handling high-dimensional, time-series and relational data features to different degrees, to handle multimodal market information for stock prediction.

- LightGBM (Ke et al., 2017): LightGBM is a gradient boosting decision tree model which is different from temporal sequence model, so that this model can only make predictions based on the input vectors.
- LSTM (Hochreiter & Schmidhuber, 1997): LSTM is an extension of RNNs that models the historical trading data of each stock and is capable of capturing the long-dependencies.
- GRU (Cho et al., 2014): GRU is a variant of LSTM which replaces the forget and input gate with only one update gate.
- GCN (Kipf & Welling, 2016): GCN is a type of convolutional neural network which has two GCN layers to propagate market information on the stock graph.
- GAT (Cheng & Li, 2021b): GAT is an extended version of GCN and consists of graph attention layers to distinguish different spillover effects for stock prediction.

2) *State-of-the-art Classifiers*: State-of-the-art models typically have well-designed modules that can address multimodal problems from a fusion perspective, thus improving stock prediction performance.

- MFNN (Long et al., 2019): MFNN is a deep learning model based on CNNs and RNNs to extract hidden features and temporal patterns to

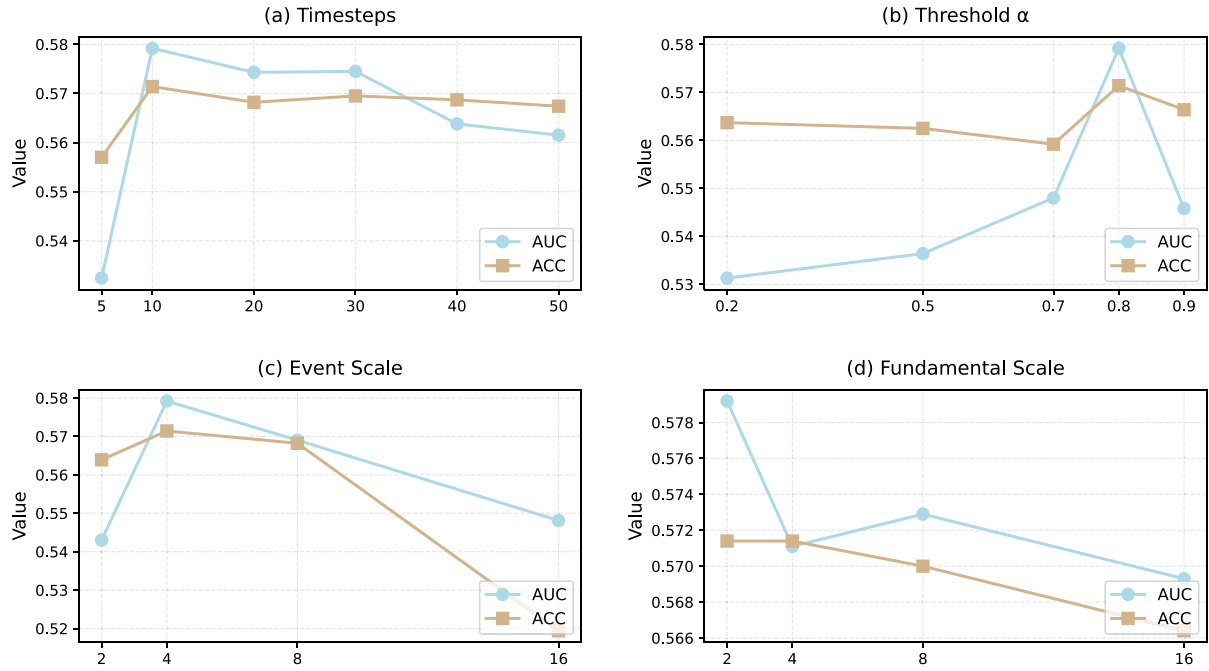


Fig. 3. The variations of DA and AUC with different parameters.

make stock predictions based on market information from multiple sources.

- MAC (Ma et al., 2023): Multi-source aggregated classification (MAC) is a recent method for stock price movement prediction which incorporates the numerical features, news sentiments of target stocks and their related stocks.
- RGCN (Li et al., 2021b): This method uses the relational GCN structure to propagate the spillover effects of related firms based on the news enhanced relationships.
- LSTM-ReGAT (Zhong et al., 2023): LSTM-ReGAT is a novel network-based model for predicting stock trends which considers extracting both sequential patterns of features and relation information. We apply our firm relationships and kept the rest of its original settings.
- StockNet (Xu & Cohen, 2018): StockNet is a deep generative model that uses tweet sentiment and historical prices to predict stock movements.
- AGATS (Zhang et al., 2025): AGATS is a dynamic attribute-driven graph attention network incorporating sentiment information, transaction data, and text data.

Table 2 presents the predictive results of baselines and the proposed multi-graph fusion framework. All methods exhibit lower performance levels on small-scale datasets compared to large-scale ones, which serves as evidence that larger-scale data is beneficial for improving generalization capability. It can be observed that LSTM and GRU, which capture the time-dependence of market information, outperform LightGBM, indicating the effectiveness of temporal information in multimodal data. When the models (RGCN, MAC and AGATS) that are capable of incorporating relational information to deal with spillover effects, are utilized, stock predictions can be further improved. This indicates that to improve the stock predictions, fusing relational information is as important as dealing with time-series financial and textual indicators. When further integrating multimodal data and incorporating both joint and spillover effects, our proposed method outperforms all the baselines with enhancements in terms of DA and AUC. The p-values from the paired t-tests are all below the significance level, indicating that the performance improvement of MGNN over the baselines is statistically significant.

Table 2

Comparison of different baselines.

Model	Small-scale		Large-scale	
	DA	AUC	DA	AUC
LightGBM	0.5375	0.5376	0.5469	0.5284
LSTM	0.5277	0.5022	0.5223	0.5538
GRU	0.5364	0.5050	0.5667	0.5652
GCN	0.5390	0.5373	0.5676	0.5592
GAT	0.5536	0.5258	0.5671	0.5515
MFNN	0.5358	0.5269	0.5467	0.5306
RGCN	0.5390	0.5373	0.5683	0.5531
LSTM-ReGAT	0.5440	0.5357	0.5675	0.5516
MAC	0.5444	0.5205	0.5653	0.5621
StockNet	0.5428	0.5387	0.5636	0.5215
AGATS	0.5632	0.5217	0.5593	0.5406
Our model	0.5637	0.5406	0.5714	0.5792

5.3. Ablation study

In this section, we conduct a series of ablation studies to examine the effectiveness of the proposed MGNN subfunctions in addressing multimodal information of stock movements on the whole dataset. Specifically, we evaluate the performance improvement brought by the event extraction of textual information, multi-graph learning and cross-attention fusion. Table 3 presents the comparison results of ablation studies in terms of DA, AUC and F1, respectively.

1) *Effectiveness of Event Extraction*: As aforementioned, addressing heterogeneity in multimodal data is salient for capturing stock movements. For this purpose, we argue that using event extraction approach to handle textual news information is a promising way to filter out noisy and redundant information and preserve valuable information. Two track experiments are conducted to examine the effectiveness of extracting event triples to quantify textual information for the subsequent fusion stage. Specifically,

- MGNN_s: This method extracts the news attributes for each firm node in terms of sentiment words as the way in the work of (Cheng & Li, 2021a).

Table 3
Comparison of different module design.

Model		DA	AUC	F1-score
News Information Extraction Method	MGNN _s	0.5350	0.5280	0.4467
	MGNN _b	0.5636	0.5513	0.5419
	MGNN _r	0.5172	0.5336	0.5132
	MGNN _{LLM}	0.5485	0.5184	0.5417
w/o Multi-graph Learning	MGNN _{F=0}	0.5727	0.5682	0.5663
	MGNN _{E=0}	0.5678	0.5609	0.5551
w/o Cross-attention Fusion	MGNN _{CA=0}	0.5635	0.5561	0.5663
	MGNN _{CA=1}	0.5506	0.5630	0.5649
	MGNN _{CA=2}	0.5658	0.5644	0.5612
Our model		0.5714	0.5792	0.5742

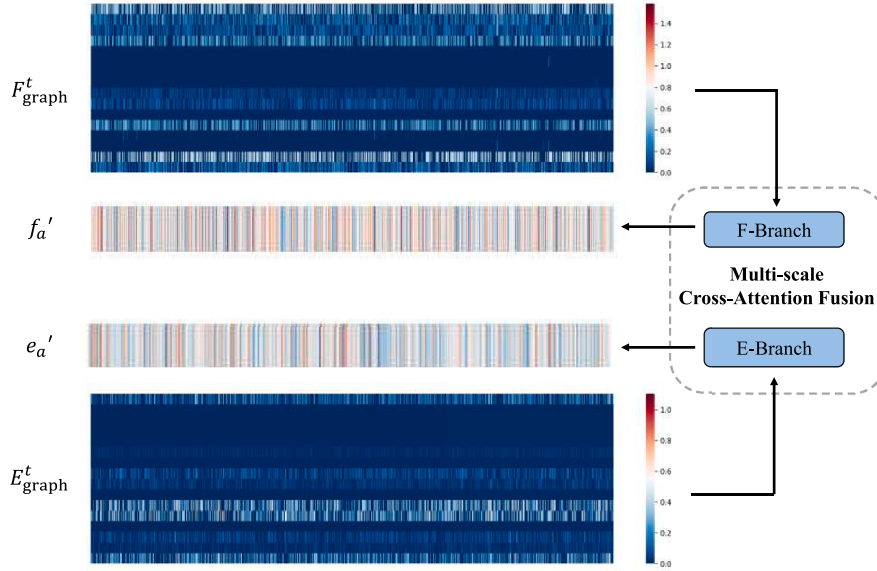


Fig. 4. A feature fusion example in multi-scale Cross-Attention fusion module.

- MGNN_b : This method is similar to MGNN_s where news attributes are represented by embedding vectors via BERT instead of sentiment words.
- MGNN_r : This method utilizes the event information encoder from REST (Xu et al., 2021b) instead of our news event extraction method.
- MGNN_{LLM} : This method utilizes LLM (Qwen3-8B Yang et al., 2025) to extract new events instead of our method.

In Table 3, it can be observed that MGNN_b method outperforms MGNN_s method, which shows that representing news with word embeddings is more efficient than the way of quantifying texts as sentiment words. The proposed MGNN approach beats both of these methods by extracting event triples to focus on influential contents. This finding consolidates our argument that simplified textual information processing will fail to find valuable information (e.g., MGNN_s) or incorporate noisy redundant information (e.g., MGNN_b). Meanwhile, the poor performance of MGNN_r and MGNN_{LLM} demonstrates that for the specialized task of financial market prediction, the predictive power of extracted events is not solely determined by the raw capability of the underlying model, but critically depends on the design of the extraction mechanism itself. Our fine-grained, structure-aware extraction method captures more actionable and economically meaningful signals than generic event extraction.

2) *Effectiveness of Multi-graph Learning*: In fact, multimodal data consists of different source data with distinct information, so a suitable fusion schema needs to be designed to combine information from data of different modalities. Apparently, incorporating spillovers from related firms requires the integration of nodal and relational information. In this study, a multi-graph learning approach is proposed to cap-

ture spillover effects by fusing relational information with fundamental and news information, respectively, to distinguish information diffusion mechanisms in different modal data. To understand the design logic, we perform a series of experiments to examine the proposed method and its alternatives. Specifically, we compared two variants of the proposed method, as follows:

- MGNN_{F=0} : This method only employs E_{graph}^t (in Section 4.2) representing information extracted from explicit event graph for stock prediction.
- MGNN_{E=0} : This method only employs F_{graph}^t (in Section 4.2) representing information extracted from implicit industry graph for stock prediction.

Table 3 indicates that removing either the implicit industry relationship or the explicit knowledge relationship would lead to a poorer prediction performance. This finding demonstrates the effectiveness of distinguishing information diffusion mechanisms in different modal data. In particular, MGNN_{F=0} outperforms MGNN_{E=0}, which further suggests that event triples extracted from textual financial news contain relational information that can improve the prediction performance. We observe that the variant MGNN_{F=0} achieves a marginally higher DA than the full model. This can be attributed to DA's relative insensitivity to predictions on the minority class in the imbalanced financial forecasting task. This result actually underscores the importance of our full fusion design.

3) *Effectiveness of Cross-attention Fusion*: Different from unimodal information, multimodal market information consists of various data structures and requires flexible fusion mechanisms to integrate all types

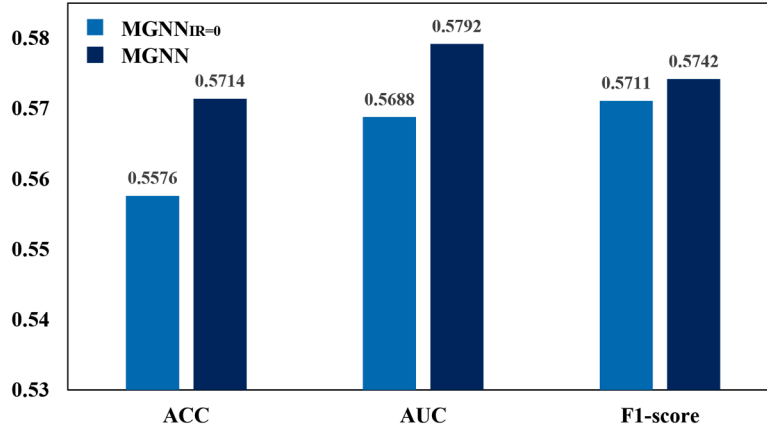


Fig. 5. Implicit relation analysis.

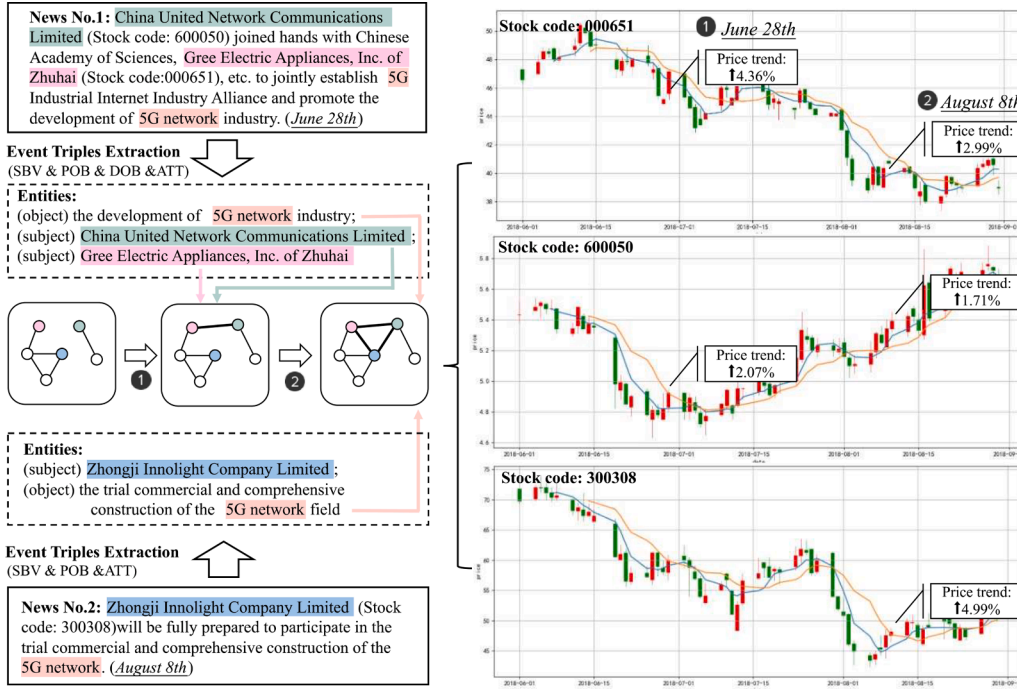


Fig. 6. Case study of the proposed MGNN.

of information in stock forecasting. A multi-scale cross-attention fusion module is proposed to further fuse the information obtained from multiple graphs to capture the joint effect of fundamental and news data. Here, we investigate the proposed fusion mechanism (Section 4.3) with two variants to understand its functionality.

- $MGNN_{CA=0}$: This method removes the cross-attention fusion module, and uses only E'_{graph} , F'_{graph} , E' , F' as the final features for stock prediction.
- $MGNN_{CA=1}$: This approach removes the multi-scale mechanism. That is, For E-Branch and F-Branch, the scales are both set as 1.
- $MGNN_{CA=2}$: This approach is similar to MGNN, but replaces the cross-attention fusion mechanism with a concatenated operation. That is, e'_a and f'_a are concatenated directly as the final features for stock prediction.

Table 3 presents that $MGNN_{CA=2}$ has a better performance than $MGNN_{CA=0}$, with the enhancements of at least 1.5% and 0.4% in terms of DA and AUC, indicating that capturing the interaction of event information and fundamental information facilitates stock forecasting. When

considering that the interactive fusion method adequately captures the joint influence of multimodal information, predictive performance of the proposed MGNN is further improved. Meanwhile, $MGNN_{CA=2}$ performances better than $MGNN_{CA=1}$. It can prove that when the data from different modal store information of varying densities, it is useful to split multi-scale segments according to the information density. To better verify this, we visualize the features in multi-scale cross-attention fusion module. Specifically, we randomly intercept the data over a certain period of time and put it into our trained model. Fig. 4 presents a visualisation of the additional tokens f'_a and e'_a for E-branch and F-branch in the form of a heatmap. In Fig. 4, it can be observed that the heatmap color of f'_a and e'_a have obvious differences, which indicates that the additional tokens store different information of input features.

5.4. Investment simulation analysis

To assess the practical utility of our proposed model in real-world financial applications, we conduct investment simulations that translate the binary classification predictions into actual Top-N trading strategies. We chose RMB 10,000 as the investment budget, and we further

Table 4
Comparative investment performance with alternative models.

Model	Top-10			Top-20		
	ARR	AVol	SR	ARR	AVol	SR
StockNet	0.1573	0.2720	0.5784	0.0708	0.2601	0.2721
AGATS	0.2105	0.2659	0.7915	0.1975	0.2585	0.7642
MAC	0.3945	0.2629	1.5004	0.2581	0.2590	0.9967
Our Model	0.5650	0.2586	2.1843	0.3535	0.2443	1.4470

assumed zero transaction costs. The experimental framework evaluates the model's ability to generate profitable trading signals and manage investment risks effectively.

For each trading day, the model generates probability distributions representing confidence scores for each prediction. Our proposed enhanced strategy selects the Top-N stocks with the highest predicted probability of rising. The capital is equally distributed among the selected stocks, where N is a hyperparameter optimized through backtesting. We employ three widely used metrics to evaluate the overall performance of each method: Annual Return Ratio (CRR), Annual Volatility (AVol), and Sharpe Ratio (SR). Lower absolute values of AVol, along with higher ARR, SR, indicate better performance. We evaluated three baselines that had performed well in previous comparative experiments under identical conditions.

The results in Table 4 reveal that the Top-10 strategy of our model significantly outperform all baseline models across all return metrics. Across all models, the Top-20 strategy produces lower returns compared to the Top-10 strategy. A higher Sharpe ratio indicates greater excess return relative to risk exposure. As presented in Table 4, which details the Sharpe ratios for both the benchmark models and our proposed method, the outcomes further substantiate the superiority of our approach. Our model maintains its performance leadership under both strategies. These findings establish that MGNN with proper investment strategy design can generate economically significant returns while maintaining reasonable risk levels.

5.5. Implicit relation analysis

The industry graph exhibits structured sparsity due to its explicit, economically defined relationships. In contrast, the knowledge graph, constructed from implicit co-occurrence and event relations in news text, demonstrates a different connectivity pattern that is denser but noisier, reflecting the broader associative nature of media-driven information. The results in Tables 3 have demonstrated that multi-graph design affect the forecasting performance. In this section, we remove the implicit relations and reserve the multi-graph structure to explore the effectiveness of implicit relations for price prediction. Specifically,

- $MGNN_{I_{R=0}}$: This method removes implicit relations and reserve the multi-graph structure, which means the two graphs are all industry graphs.

Fig. 5 represents the implicit relation analysis. In Fig. 5, it can be observed that the prediction result of multi-graph learning with the implicit relations performances better than that of multi-graph learning with two same explicit graphs with an enhancement of 1.38, 1.04 and 0.31 percent. This indicating that implicit relations extracted from financial news contain useful information for stock predictions.

6. Case study

This section describes a case study for a better understanding of the proposed MGNN. Here, we take China Unicom (600050) as our example. It is the third-largest wireless carrier in China and the sixth largest mobile provider in the world. This firm currently provides a range of services including mobile network, long-distance, data communications,

and Internet services. In Fig. 6, a news article reports that there is a collaboration between China Unicom and Gree Electric (000651) in the 5G field. Although China Unicom and Gree are not in the same industry and do not share a common major shareholder, our event extraction methodology can identify their relationship from this firm-specific news. In addition, the second piece of news indicates that Zhongji Innolight will be fully prepared to participate in 5G construction. With the proposed fine-grained event extraction, the entity 5G network can be effectively identified and connect China Unicom, Gree and Zhongji by covering the same entity in the triples. Based on the enhanced relationship, the price movements of the three related stocks show similar temporal trends and similar detailed fluctuations, which collectively result in an approximate 2.1 percent rise in stock prices. This finding demonstrates the effectiveness of the proposed event extraction method and the importance of spillover effects in stock predictions.

Based on the aforementioned facts, it can be inferred that the extracted event triples show potential relations between firms and have considerable usefulness in capturing media spillovers.

7. Conclusion

The stock market is a complex dynamic system where various market factors mingle together to cause stock fluctuations. It is of great necessity to incorporate multimodal market information for forecasting stock movements. To reach this goal, the heterogeneity of multimodal data and their interactions are two main concerns in the design of fusion schema. In this study, we propose a multi-graph learning framework for tracking stock trends in term of multimodal market information. In particular, a fine-grained event extraction approach is utilized to process textual news. The obtained information reflecting valuable events at the company level is of high quality and has little interfering information, which allows for better subsequent integration. In addition, we fuse the explicit firm relationship with fundamental information to capture momentum spillovers and use the implicit firm relationship based on extracted event triples to incorporate media spillovers. With such a multi-graph design, multimodal fundamental information and financial news at the firm level can be fully integrated with firm relationships at the interfirm level, respectively. To fully integrate the multimodal information, a cross-attention dual-branch transformer is proposed to capture the joint influence of different modes on stock movements. The experimental findings of this study consolidate the effectiveness of the proposed framework for addressing the heterogeneity of multimodal data and their interactions. Additionally, many other domains, including social recommendations, financial derivative investments and visual question answering, follow this vein when multimodal information exists. More relevant studies are expected to be conducted in the near future.

CRedit authorship contribution statement

Zhixi Li: Conceptualization, Methodology, Software, Writing – original draft, Visualization; **Jie Xiong**: Conceptualization, Methodology, Software, Visualization; **Jun Wang**: Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing; **Jinghua Tan**: Conceptualization, Methodology, Writing - original draft, Writing – review & editing; **Philippe Du Jardin**: Conceptualization, Methodology; **Muhammet Deveci**: Conceptualization, Methodology; **Kaiyang Zhong**: Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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